

AMENDMENT NO. 1 NOVEMBER 2022

TO

IS 12837 : 1989 HYDRAULIC TURBINES FOR MEDIUM AND LARGE POWER HOUSES — GUIDELINES FOR SELECTION

(Page 1, clause 3.1) — Substitute the following for the existing clause:

‘3.1 Maximum Net Head

The maximum net head available for power generation is difference in maximum forebay level and tail water level with one unit operating at no load (turbine discharge about 5% of rated flow) less all losses for water conductor system including penstock.’

(Page 1, clause 3.8) — Insert the following clauses:

‘3.8 Rated Head

Net head at which full gate output of the turbine produces the generator rated output in kilowatts.

3.9 Design Head

Net head at which peak efficiency is desired.’

(Page 2, clause 5.1, Table 1) — Insert the following table for the existing:

Table 1 Criteria for Selection of Hydraulic Turbine

(Clause 5.1)

Sl No.	Type of Machine	Head Variation Percent of Rated Load	Load Variation Percent of Rated Output	Specific Speed	Weighted Average Efficiency (Percent)
(1)	(2)	(3)	(4)	(5)	(6)
i)	Pelton	120 – 80	50 – 100	15 – 65	92
ii)	Francis	125 – 65	50 – 100	60 – 400	95

Table 1 (Concluded)

Sl No.	Type of Machine	Head Variation Percent of Rated Load	Load Variation Percent of Rated Output	Specific Speed	Weighted Average Efficiency (Percent)
(1)	(2)	(3)	(4)	(5)	(6)
iii)	Deriaz	125 – 65	50 – 100	200 – 400	94
iv)	Kaplan	125 – 65	40 – 100	300 – 800	94
v)	Propeller	110 – 90	90 – 100	300 – 800	94
vi)	Bulb	125 – 65	40 – 100	600 – 1200	94

(Page 2, clause 5.3) — Add the following below the table:

‘NOTE — For short duration, recommendation of manufacturer may be sought in the specifications.’

(Page 2, clause 5.5, sentence 4) — Substitute the following for the existing sentence:

‘In case of Francis turbines, necessity of limiting runner size during transport, requires manufacture of runner in two pieces.’

(Page 3, clause 5.6, informal table) — Substitute the following for exiting informal table

Type of Turbine	Pressure Rise (Percent)	Speed Rise (Percent)
Reaction	< 35	< 60
Pelton	< 15	< 20